Contact App

Maintain a private contact directory with create, view, edit, delete and search, a custom RecyclerView adapter and local SQLite storage.

Requirement coverage

Contact model and validation - A contact has a required name and phone, with optional email. Names allow 1-120 characters; phone numbers need 3-20 digits with basic formatting, up to 40 characters.

Directory and CRUD - SQLite persists entries. A custom ListAdapter displays names and phones in alphabetical order. Search matches name, phone or email without treating SQL wildcards specially.

Viewing and dialer - Selecting a row shows full contact details. Open dialer uses ACTION_DIAL with the selected number and handles missing dialer apps. The app requests no contacts or call permission.

Run and review

Open the source folder in Android Studio, install SDK 35 / Build Tools 35.0.0 and use JDK 17 or 21. Sync Gradle and run the contacts module on Android 8.0 / API 26 or later.

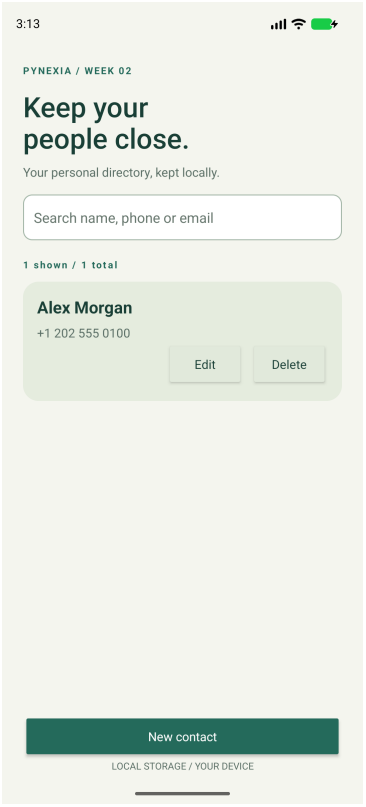
1. Tap New contact and enter a name, phone and optional email.
2. Save, then search by name, phone digits or email.
3. Open the contact details and select Open dialer to inspect the number.
4. Edit the entry, reopen the app, and test cancelling/confirming deletion.

Scope

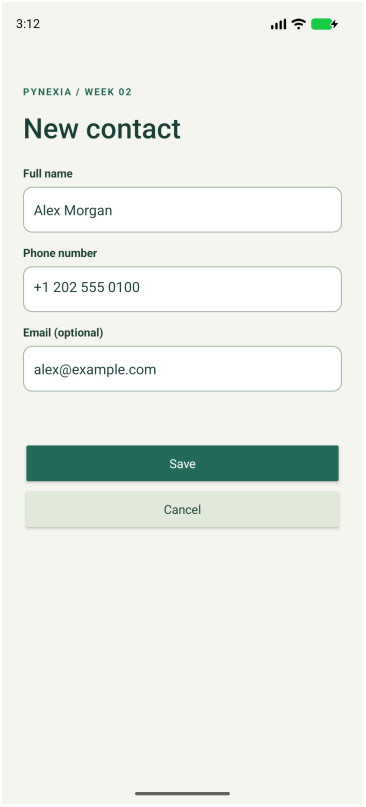
Does not read or modify the system address book. No automatic calls, cloud sync or encrypted storage. Clearing app data/uninstalling removes contacts. Phone validation is a basic input check, not number verification.

Actual app screens

Captured from the tested native app on the emulator.



Local contact directory



Contact editor

Executed verification

4 JVM tests + 4 device tests passed. Lint: 0 errors, 18 warnings. No failures, errors or skipped tests.

Name/phone/email validation and literal search, SQLite CRUD/reopen, UI add/edit/delete, editor recreation, full details and intercepted ACTION_DIAL Intent.

Pixel 8 emulator, Android 17 / API 37 (37.1 image), x86_64. No physical-device or API 26 test run. Lint warnings are retained without suppression. APKs are debug-signed evaluation builds.

Reproduce

.\gradlew.bat :contacts:assembleDebug :contacts:testDebugUnitTest :contacts:lintDebug

.\gradlew.bat :contacts:connectedDebugAndroidTest

The second command needs a running emulator or connected device. Initial Gradle sync needs internet. See verification/ for executed test results and lint findings.

Explanation & submission

يحفظ التطبيق جهات الاتصال في دليل محلي مستقل عن جهات اتصال الهاتف. يمكن إضافة الاسم ورقم الهاتف والبريد الاختياري وعرض البيانات وتعديلها وحذفها. يبحث في الحقول الثلاثة ويرتب الأسماء، ويفتح برنامج الاتصال بالرقم المحدد دون إجراء مكالمة تلقائيا.

Repository URL

https://github.com/ENGEyad/pynexia-week2-contacts

Live Demo / Hosted URL

https://engeyad.github.io/pynexia-week2-contacts/

A hosted project showcase with real screenshots and downloads, not an Android emulator.

Attach File / Archive

Contact-App.zip - source, Gradle wrapper, APK, screenshots, verification and this PDF. Below the 10 MB limit. APK is inside the ZIP because the earlier submission form does not list direct APK uploads.

Notes / Implementation Remarks (1422 / 2000 characters)

Native Android Contact App implemented with Java and XML. Maintain a private contact directory with create, view, edit, delete and search, a custom RecyclerView adapter and local SQLite storage.

A contact has a required name and phone, with optional email. Names allow 1-120 characters; phone numbers need 3-20 digits with basic formatting, up to 40 characters. SQLite persists entries. A custom ListAdapter displays names and phones in alphabetical order. Search matches name, phone or email without treating SQL wildcards specially. Selecting a row shows full contact details. Open dialer uses ACTION_DIAL with the selected number and handles missing dialer apps. The app requests no contacts or call permission.

Run: open the source folder in Android Studio, sync Gradle, and run contacts on Android 8.0/API 26 or later (compile/target SDK 35).

Validation: 4 JVM tests and 4 device tests passed on a Pixel 8 emulator, Android 17/API 37. Lint: zero errors; 18 warnings documented without suppression.

Does not read or modify the system address book. No automatic calls, cloud sync or encrypted storage. Clearing app data/uninstalling removes contacts. Phone validation is a basic input check, not number verification.

Hosted URL: project showcase with actual screenshots and downloads, not a browser emulator. The ZIP contains source, Gradle wrapper, debug APK, screenshots, verification and a bilingual PDF report.